LESSON PLAN

AUXILIARY EQUIPMENT (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and two stroke Piston engine, comparison, advantages, modified Otto cycle
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	IC Engines, Variations and usage in aviation industry as main propulsion plant and auxiliary machinery, operation cycles of four stroke and two stroke Piston engine, comparison, advantages, modified Otto cycle
5. Specific Objectives	22.1.1.1 Trainees will be able to explain about the working principle of piston engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine cutaway diagrams; Safety: Handle diagrams carefully; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an internal combustion engine?
(b) Name two types of piston engines used in aviation.
(c) Define the term 'propulsion plant'.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Working principle of piston engines • Four stroke cycle operation • Two stroke cycle operation • Comparison of four and two stroke engines	L & Ds	PPT, IB	15 min
(b)	• Usage in aviation main propulsion • Usage in auxiliary machinery • Advantages of piston engines • Modified Otto cycle explanation	L & Ds	PPT, PDF	20 min
(c)	• Detailed comparison of cycles • Efficiency analysis • Application scenarios • Summary of key differences	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between the four stroke and two stroke cycles.
(b) Describe the role of the modified Otto cycle in aviation applications.

12. Home Assignment and Follow Up	Prepare a brief note on how superchargers modify the operation of piston engines, linking to the principles covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principles & operations of Super chargers, description and operations	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min